

TIC 239661046

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 1.053991 d
Epoch (t0) 1816.9715 BTJD
Duration 0.0580 d
Depth 6.249 % (62,488 ppm)
SNR 889.2

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

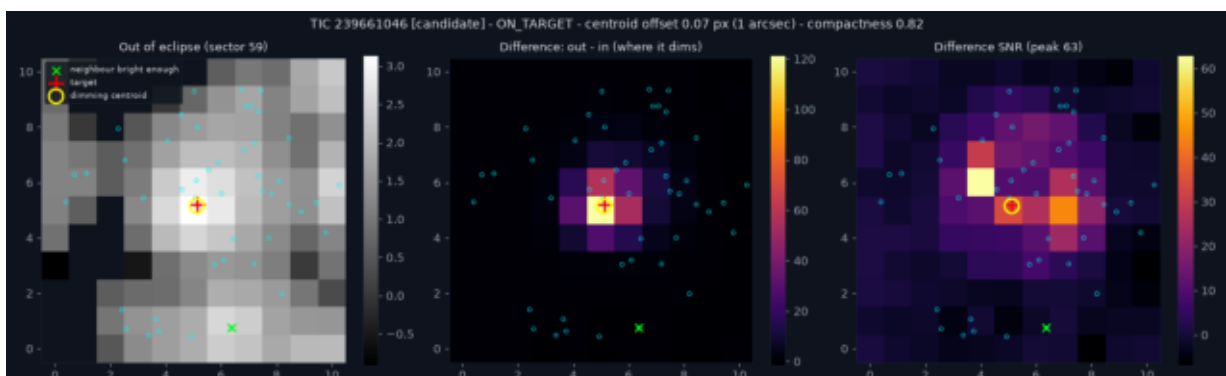
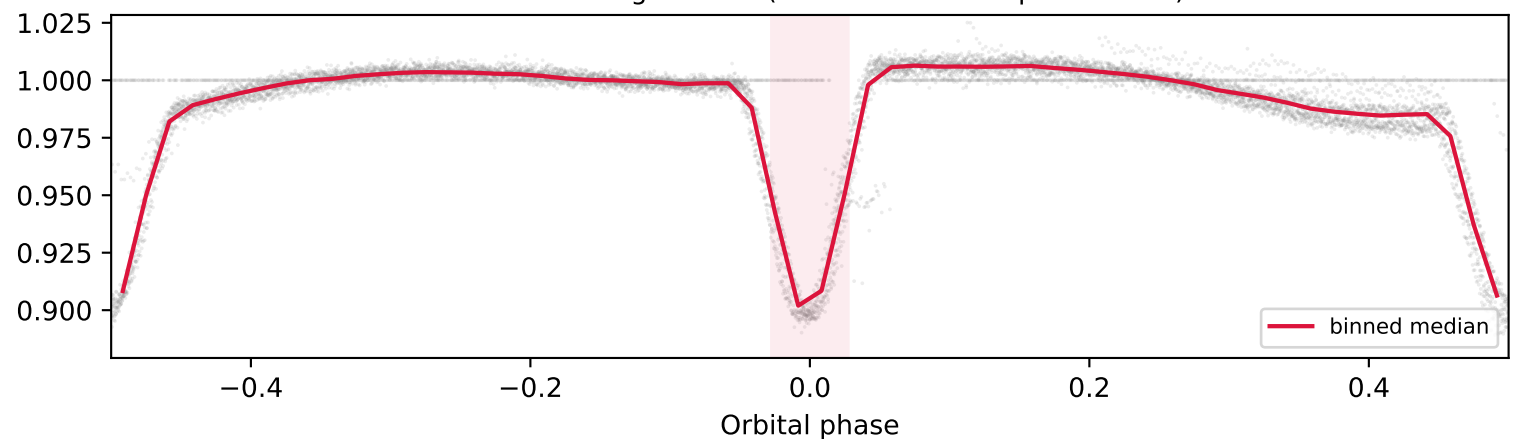
Eclipses observed 36 (gate ≥ 3)
Robust depth ratio 1.32 (gate ≥ 0.50)
Robust depth sigma 248.6 (gate ≥ 7.0)
Odd/even z-score 0.50 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 59 light-curve dip in this sector: 409.3 sigma t0 refined by -0.0025 phase
Difference-image peak SNR 63.02 (gate ≥ 5.0) Compactness 0.82 (gate ≥ 0.50)
Dimming centroid offset 0.07 px (1 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 239661034
4.57 px away, dTmag +1.39 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.